

Licensing

Use a **VCF Operations** instance and the **VCF Business Services** console as a VI administrator to manage and assign licenses.

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Getting to Know VCF

To apply this documentation, you must be acquainted with the [VCF Product Overview](#), [VCF Design](#), and [VCF Release Notes](#).

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Licensing Overview

Starting with version 9.0 of VMware Cloud Foundation (VCF) and VMware vSphere Foundation (vSphere Foundation), you license your environment by using a VCF Operations instance and the VMware Cloud Foundation Business Services console ([vcf.broadcom.com](#)). Subscription-based license files replace the use of the 25-character license keys.

What's New in 9.0?

- License keys are no longer used. To license your environment, you purchase a term-based subscription for a product, with a specified capacity.

- A license is an object that entitles you to use the products that you purchased subscriptions for. A license can span multiple subscriptions or, when you split the default license, a license can be a portion of a subscription.
- The initial license that is automatically created in VCF Business Services console after you purchase a subscription is called a default license. The default license is a pool of capacity from all active subscriptions of the same product in the same Site ID. Available capacity is displayed in the unit of measure for the specific product.
For example, if you purchase a VCF subscription for 500 cores, you receive a default license for 500 cores of VCF and you also receive a default license of 500 TiBs of vSAN, which are added in a separate license. If, on top of that, you purchase a vSAN add-on subscription for 200 TiBs, the new number of TiBs is added to your default vSAN license, and its capacity becomes 700 TiBs.

Note:

For more information about using your Broadcom Site ID, see [knowledge base article 142873](#).

- There are two types of licenses - primary licenses, such as VMware Cloud Foundation and VMware vSphere Foundation licenses, and add-on licenses, such as vSAN add-on capacity or VMware Private AI Foundation with NVIDIA licenses. You no longer license individual components such as NSX, HCX, VCF Automation, and so on. Instead, for VCF and vSphere Foundation, you have a single license capacity provided for that product.
For example, for VMware Cloud Foundation, the single license capacity provided for that product is *VMware Cloud Foundation (cores)*. This is a primary license and you must assign it to your environment before you can assign an add-on license. Components are licensed automatically after you assign the primary license to a vCenter instance. For VCF and vSphere Foundation, you also receive *VMware vSAN (TiB) capacity*. You also receive *VMware vSAN (TiB) capacity* when you purchase vSAN as an add-on. The vSAN TiB license is called an add-on license.
- In VCF 9.0, the two supported add-on licenses using the new licensing system are *VMware vSAN (TiB)* and *VMware Private AI Foundation with NVIDIA (cores)*.
- To license your environment, you must have an installed VCF Operations instance, register it in the VCF Business Services console, and add licenses with available capacity. You can register VCF Operations to the VCF Business Services console either in connected or in disconnected mode. Internet connection is not required to register but connected registration is recommended because it simplifies license management. You assign licenses only to the vCenter instances. The other product components added to the licensed vCenter instances are then automatically licensed. For more information about products and product components, see [Products and Product Components](#).
- License usage reports are required at least once every 180 days to maintain your licenses and you must update your license to confirm that the license usage report was submitted. If VCF Operations is registered in connected mode, this data is sent to the VCF Business Services console automatically, and licenses can be updated with a button click. If VCF Operations is registered in disconnected mode, to report license usage, you generate a usage file and upload it in the VCF Business Services console ([vcf.broadcom.com](#)). For detailed instructions for both connected and disconnected registration modes, see [Updating Licenses](#).
- License expiration applies if the subscription expires or if usage is not reported and the license is not refreshed:
 - Notifications are displayed before the license expiration.
 - After the license expires, you have 90 days to update your license. License assignments are impacted during this 90 day period and notifications appear in most components, including VCF Operations and vCenter.
 - 90 days after the expiration, your environment is impacted in several ways, including:
 - Management operations of various components might be prevented;
 - ESX hosts disconnect from vCenter, most host management operations are prevented;
 - You cannot create or start workloads.

Note:

Existing workloads are not proactively stopped.

- You can recover expired environments by downloading an updated license.

- With your first subscription for a product, you receive both a version 9.0 default license for the product, and license keys for version 8.x of the product. You can decide which version of the product to deploy and license, but your total usage must not exceed your purchased capacity. For more information, see [Products and Product Components](#). The

capacity that you receive with your next subscriptions for the same product is added to the default license for this product.

- You can check your license consumption analytics in your VCF Operations instance, and the VCF Business Services console. For more information about usage analytics, see [License Usage Analytics](#).

Licensing Terminology

As part of the licensing workflows, specific terminology is used to refer to different licensing-related objects and actions.

Term	Related actions	Description
Subscription		You purchase a subscription for a specific product. You decide what capacity to purchase for the subscription. Subscriptions are term based.
Assets		Any object in your environment that requires licensing. Assets are vCenter instances, ESX hosts, VCF Operations instances, NSX, and others. Starting with 9.0, you assign a license to a vCenter instance, and all other assets, which are connected to that vCenter instance, are licensed automatically.
License		An object that entitles you to use the products that you purchased subscriptions for. A license can span multiple subscriptions or, when you split the default license, a license can be a portion of a subscription.
	Create/Split	Performed in the VCF Business Services console. The process of generating a new license. You allocate to it a portion from the capacity of a default license. For more information, see Split Default Licenses .
	Add (to VCF Operations)	Performed in the VCF Business Services console. The process of associating the license with a VCF Operations instance. Then you can assign that license to vCenter instances from the VCF Operations instance. For more information, see Add a License to VCF Operations .
	Remove (from VCF Operations)	Performed in the VCF Business Services console. When you remove a license from a VCF Operations instance, the license is no longer available in that VCF Operations instance, and you can add it to a different VCF Operations instance. For more information, see Remove a License from VCF Operations .
	Delete	Performed in the VCF Business Services console. After you delete a license, the license is no longer available and the allocated capacity from this license is automatically merged into the capacity of the corresponding default license. For more information, see Delete a License .

	Assign	Performed in VCF Operations. The action of entitling a vCenter instance, the cores or TiBs on assets within the vCenter instance, including hosts or vSAN clusters, and any connected component, for example NSX. For more information, see Assign a Primary License to a vCenter Instance or a VCF Instance and Assign an Add-on License to a vCenter Instance .
Default license		The initial license that is automatically created in the VCF Business Services console after you purchase a subscription. The default license is a pool of capacity from all active subscriptions of the same product in the same site.
Split License		A license that you create manually. You allocate to it a portion from the capacity of a default license. For more information, see Split Default Licenses .
Primary license		VCF and vSphere Foundation licenses.
Add-on license		You purchase them separately from the primary license, or, in the case of vSAN, you also receive capacity for the add-on license with your primary product purchase. You can assign add-on licenses to vCenter instances only after you assign a primary license. For example, in VCF 9.0 and vSphere Foundation 9.0, two of the supported add-on licenses are <i>VMware vSAN (TiB)</i> and <i>VMware Private AI Foundation with NVIDIA (cores)</i> .
License file		The license file contains information that proves to VCF Operations that your environment is licensed. A license file can contain multiple licenses.
License Unit		The unit of measure for capacity, for example cores and TiBs.
Capacity		The number of cores or vSAN storage TiBs.
	Allocate	The process of adding cores quantity or vSAN storage capacity to a license through splitting or merging.
Allocated capacity		A portion of the default license capacity that you added to a new license. Allocated capacity can be measured in cores or TiBs.
	Increase	The capacity allocated to a license can be increased up to the available capacity in the default license. For more information, see Change License Capacity .
	Decrease	The capacity allocated to a license can be decreased. For more information, see Change License Capacity .

Total capacity		The total capacity of a product equals the combined capacity of the default license and all other licenses split from that default license.
Used capacity		Per license, the actual capacity consumed by the assets in your environment.
Unused capacity		Per license, the part of the allocated capacity that is not yet consumed by assets in the vCenter instances to which the license is assigned.
Over use		When a license has more used capacity than its currently allocated capacity.
Pooling		Combining the total capacity of subscriptions for the same product in one license.

9.0 Product Subscription and Licensing

VMware Cloud Foundation and vSphere Foundation include different sets of product components. You purchase a subscription for a specific product. With the provided default primary license, you license all product components by using a registered VCF Operations instance.

For example, when you purchase a subscription for VCF, with the license you receive and assign to a vCenter instance, all components connected to that vCenter instance are licensed automatically. With your subscription for a product, you also receive license keys for the earlier version of the product and product components. You can decide which version of the product to deploy and license, but your total license usage must not exceed your purchased subscription capacity.

Purchased Product	Pre-Version 9 License Key Examples	Version 9 License Equivalent
VMware Cloud Foundation (VCF)	- VMware Aria Operations Networks - VMware Aria Suite Enterprise 8.14 for Cloud Foundation - VMware Cloud Director - VMware HCX Enterprise - VMware HCX Advanced - VMware NSX Networking for Cloud Foundation - VMware SDDC Manager 5.x for Cloud Foundation - VMware vCenter Server 8 Standard Term License per Instance - VMware vSphere 8 Enterprise Plus for Cloud Foundation (Supports vCenter Server 8.0.0a and above) - VMware Tanzu Kubernetes Grid for vSphere Foundation	VMware Cloud Foundation (cores)
	VMware vSAN 8 for Foundation (Supports vCenter Server 8.0u3 and above)	VMware vSAN (TiB)

Additionally, you can purchase subscriptions for add-ons which are compatible with your products. For information about available add-ons for each product, contact your sales team. Add-ons do not necessarily follow the licensing model that VCF and vSphere Foundation have. You can find information about how to license a specific add-on within their respective documentation.

Important:

To use VCF Operations for logs, you must integrate it with a vCenter instance of version 9 that is licensed with a VMware vSphere Foundation license or a VMware Cloud Foundation license. At least one vCenter integration must be configured

to collect logs directly into a VCF Operations for logs cluster, as opposed to collecting logs by using a collector or a collector group.

Licensing Model

Starting with version 9.0, the licensing model is the same for VCF and vSphere Foundation. You assign licenses only to vCenter instances. The other product components, including ESXi hosts, that are connected to the licensed vCenter instances, are licensed automatically.

How Capacity Pooling Works

When you purchase multiple subscriptions for the same product, capacity is pooled together in one default license per product. For example, if you purchase a subscription for 200 cores for VCF, and then you purchase another subscription for 300 cores, you have one default license for the product with 500 cores total capacity, instead of two licenses with 200 and 300 cores capacity. If the different subscriptions have different end dates, the number of cores will be reduced on the end date of the subscription, but you can use the capacity for which you have an active subscription.

vSAN capacity, when purchased through VCF, vSphere Foundation, or a vSAN add-on also pools together in one default license. For example, when you purchase a subscription for 300 cores for VCF, and 50 TiBs as a vSAN add-on, you have one default license with 350 TiBs total capacity.

To pool together into the same default license, subscriptions must meet the following requirements:

- Must be for the same product, for example, VCF, or vSphere Foundation.
- Must have the same unit of measure, for example, cores or TiBs.
- Must have the same license (Site ID).
- Must not support overage billing. VMware Cloud Services Provider (VCSP) partner subscriptions are not eligible for pooling.

Note:

For some products, you receive multiple default licenses with one subscription. For example, VCF has both a default VMware Cloud Foundation license and a vSAN Enterprise license.

What Are Primary Licenses

Primary licenses are the licenses for VCF and vSphere Foundation. They are per-core based. The only assets in your environment which consume the capacity of your primary licenses are ESX hosts. To calculate the capacity you need for your environment, you need the total number of the physical CPU cores for each physical CPU on all ESX hosts in your environment. Most products have a minimum consumption rule of 16 cores per physical CPU. CPUs that have fewer than 16 cores consume 16 cores. For example, if you have 1 ESX host in your inventory, with 1 CPU, and 8 CPU cores per CPU, this ESX host will use 16 cores of your license total capacity because it is the minimum license capacity.

Number of ESX Hosts	Number of CPUs per Host	Cores per CPU	Required License Capacity
1	1	8	16
2	2	8	64
2	2	16	64
2	2	24	96

For more information about calculating the license capacity you need for your environment, see Broadcom knowledge base article [95927](#).

You assign primary licenses to your vCenter instances. The other components of the product are licensed automatically with the license that you assign to the vCenter instance. For more information about the products and their components, see [Products and Product Components](#).

To assign licenses to your vCenter instances, you use a VCF Operations instance. For more information about the licensing workflow, see [High-Level Licensing Workflow](#).

Note:

The difference in the licensing model for VCF Edge and VMware Cloud Foundation is that VCF Edge has an 8 CPU cores per physical CPU minimum, and deployment limitations. For more information about VCF Edge license terms, see the [Specific Program Documentation](#) for VCF Edge.

Evaluation Mode

After you install a product or component, it operates in evaluation mode for up to 90 days. During that period, you must license your environment.

Note:

ESX hosts that you provision as stateless hosts by using Auto Deploy have no evaluation period in version 9.0.

High-Level Licensing Workflow

1. Install a VCF Operations instance version 9.0, or upgrade your existing Aria Operations instance to VCF Operations 9.0.

Note:

VCF Operations works in evaluation mode for up to 90 days after you upgrade it from version 8. You can complete the remaining steps during this period.

2. Register your VCF Operations instance with the VCF Business Services console by using one of the two modes:
 - Disconnected mode - no Internet connection is required. You must manually exchange registration and license file between the VCF Operations instance and the VCF Business Services console.
 - Connected mode - Internet connection is required. You use an activation code from the VCF Business Services console to register your VCF Operations instance.
3. Add licenses to the VCF Operations instance from the VCF Business Services console.
4. Assign licenses to vCenter instances from VCF Operations.

Figure 124: High-Level Workflow Diagram for Licensing

VCF Business Services console
vcf.broadcom.com

Important:

After you add licenses to VCF Operations, you must update the added licenses at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations. Existing workloads are not proactively stopped.

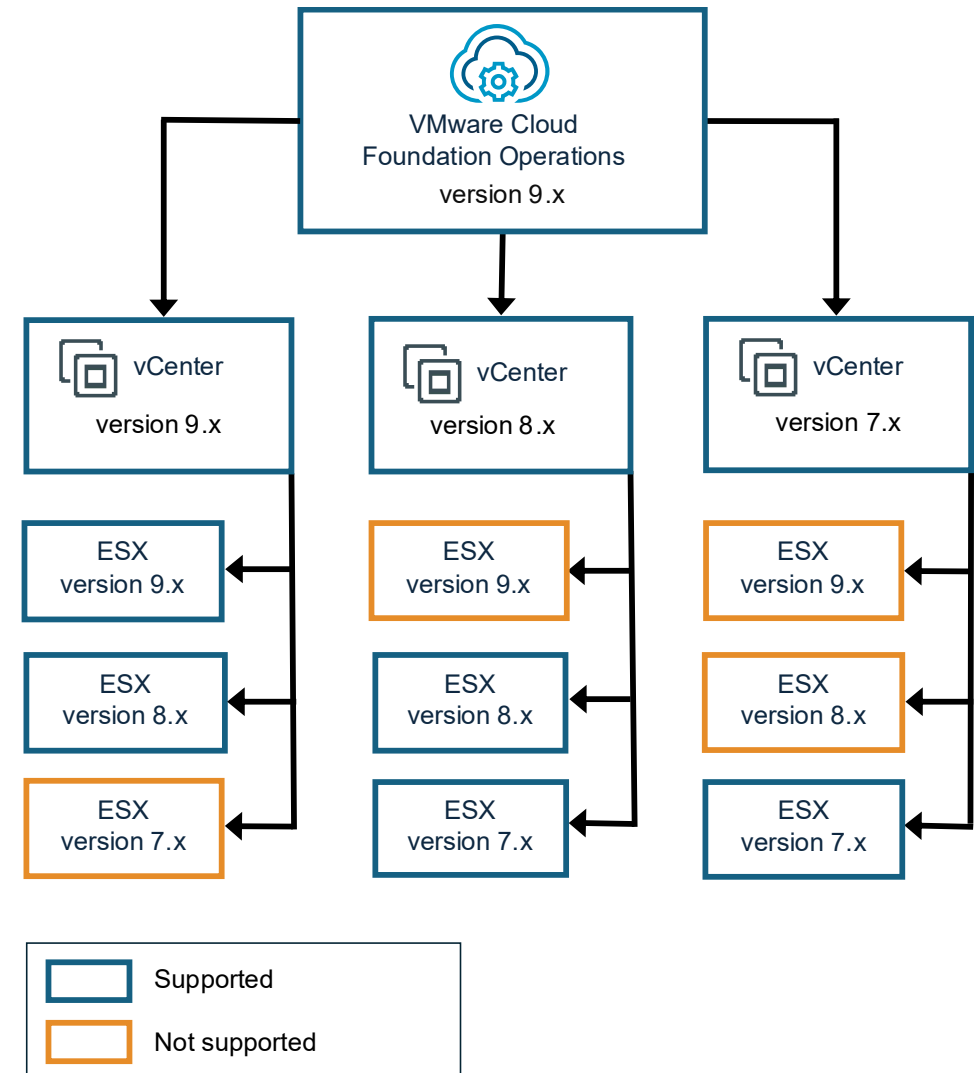
For more information, see [Update Licenses](#).

Mixed-Version Environments

If you have pre-version 9.0 product components, for example vCenter instances, ESX hosts, NSX, you can continue to use license keys to license capabilities of those components without any changes. If you use a VCF Operations instance for license management, you must update the instance to manage version 9+ licenses.

You can add ESX 8.x hosts that are licensed with license keys to your vCenter instances of version 9. Those ESX hosts must be licensed with a version 8 license. Select a license that has the same feature list as your version 9 license.

In a VMware Cloud Foundation Instance, the management domain of version 9.0 can manage workload domains of version 5.x and 9.0.

Figure 125: Example of Mixed-Version Environment

License usage data is collected for version 9 licenses, and for licenses of earlier versions connected to VCF Operations. You can view both your version 9 and pre-version 9 license usage data by using the usage analytics available in VCF Operations and the VCF Business Services console.

When you upgrade your vCenter instances to version 9, they operate in evaluation mode for up to 90 days. During that period, you must license the vCenter instances with a primary license of version 9+ with sufficient capacity. The license must be for the product you want to use.

vSAN Add-on License

To be able to use an add-on license, you must first assign a primary license to your vCenter instances. Add-on licenses are available for VCF and vSphere Foundation products.

This documentation includes detailed information only about the vSAN add-on. For a full list of add-on licenses for your product, contact your sales team. For licensing information for specific add-ons read their respective documentation.

vSAN Licensing Overview

When you purchase VCF, in addition to your VMware Cloud Foundation (cores) license you receive an equivalent vSAN entitlement for terabytes (TiB) of total raw physical storage. You can also purchase additional storage as a separate add-on license.

What's New?

- You can assign a vSAN TiB license to a single vCenter or multiple vCenter instance.
- vSAN license usage is based on the storage capacity in the vSAN clusters. vSAN pools the capacity of the subscriptions that you purchased and shares the allocated capacity between multiple vSAN clusters using the vSAN TiB license.
- You do not need a separate license or subscription for vSAN witness. You can deploy as many vSAN witness appliances as required. vSAN witness may not be used to run workloads not related to vSAN witness.

vSAN TiB License

When you assign a vSAN TiB license to vCenter, the vSAN license is automatically applied to all the vSAN clusters that are managed by the vCenter. To calculate the capacity that you need for your vSAN environment, you must determine enough TiB licenses required for the total physical device capacity in TiBs on all the ESX hosts in each vSAN cluster.

The vSAN clusters reflect the total raw storage capacity available. For more information about calculating the license capacity that you need for your environment, see the Broadcom knowledge base article [95927](#).

If you have a bare metal host as a vSAN witness, perform the following:

- Log in to the witness host.
- Run the following command:

```
esxcli vsan witness license set --enable true
```
- Reboot the witness host.

vSAN License Capacity in VMware Cloud Foundation

If you have a vSAN cluster with 3 ESX hosts, 1 CPU per host, and each host has 6 physical cores per CPU, you must purchase the subscription capacity of 16 cores per CPU as it is the required minimum license capacity. For each VCF cores that you purchase, you receive 1 TiB of vSAN capacity. With a total of 48 (3 * 1 * 16) physical cores per VCF cluster, you receive 48 TiB vSAN capacity. Similarly, with a total of 144 (3 * 2 * 24) physical cores per VCF cluster, you receive 144 TiB vSAN capacity.

Number of ESX hosts (in vSAN cluster)	CPUs per ESX Host	Cores per CPU	Number of Core License	Entitled vSAN Capacity (TiB)
3	1	6 (minimum 16 cores required)	48	48
3	2	24	144	144

You need to purchase vSAN add-on licenses if you need additional capacity. For capacity larger than the total entitled vSAN capacity in terabytes, you can purchase additional vSAN capacity. When you purchase additional capacity, you receive a vSAN license.

The license use of vSAN is recalculated and updated in the following cases:

- If you assign a new license to the vSAN cluster
- If you add a new host to the vSAN cluster
- If a host is removed from the cluster
- If the total number of TiBs in a cluster changes

vSAN License Capacity in VMware vSphere Foundation

With the VMware vSphere Foundation license, you receive 0.25 terabyte (TiB) of vSAN storage per vSAN host physical core.

To calculate the capacity that you need for your vSAN environment, you need the total number of licensed CPU cores for each CPU on all the ESX hosts in your environment. For example, consider a vSAN cluster with 3 ESX hosts, 1 CPU per host, and each host has 8 physical cores per CPU. You can use up to 0.25 TiB of included vSAN storage per vSAN host physical core. You must purchase a vSphere Foundation license with the subscription capacity of 16 cores per CPU because it is the required minimum license capacity. With a total of 48 (3 * 1 * 16) physical cores per CPU, you receive 12 TiBs (0.25 TiB * total cores in the vSAN cluster) capacity. Similarly, With a total of 128 (4 * 2 * 16) physical cores per CPU, you receive 32 TiBs capacity.

Number of ESX hosts (in vSAN cluster)	CPUs per ESX Host	Cores per CPU	Number of Core License	Entitled vSAN Capacity (TiB)
3	1	8 (minimum 16 cores required)	48	12
4	2	16	128	32

The 0.25 Tib per vSAN host physical core of license capacity pools with any vSAN capacity from any other add-on or VCF purchases. You can use this capacity to license the vSAN clusters in their vCenter instances, regardless of whether the capacity is more or less than 0.25 Tib per core.

Registering VCF Operations with the VCF Business Services console

Starting with version 9.0, you access and manage the licenses for your environment by using VCF Operations and the VCF Business Services console. You must register the VCF Operations instance with the VCF Business Services console.

You must install a VCF Operations instance version 9.0 in order to access and manage your licenses. If you want to access and manage your 9.0 licenses, and you already have an earlier version of VCF Operations installed, you must upgrade the instance to version 9.0.

Note:

If you want to use a VCF Operations instance only for monitoring your assets, and not for license management, you do not need to register it with the VCF Business Services console. If used for monitoring purposes only, the VCF Operations

instance is automatically licensed with the license assigned to the vCenter instances of version 9 added to it if those vCenter instances are licensed from another VCF Operations instance.

After you deploy or upgrade your VCF Operations instance to version 9.0, it works in evaluation mode for up to 90 days. During that period you must register VCF Operations in the VCF Business Services console and license your environment.

You can use specific VCF Operations capabilities based on the primary license that is assigned to each vCenter.

Product	VCF Operations Capabilities
VCF	License Management, Monitoring, Fleet Management
vSphere Foundation	License management, Monitoring

One VCF Operations instance can manage a mix of version 9.0 licenses.

You can choose whether to register your VCF Operations instance in connected or disconnected mode.

- In disconnected mode, to register VCF Operations, you must manually transfer files between the VCF Operations instance and the VCF Business Services console. After the registration, you must manually transfer files to report license usage, and to update licenses.
- In connected mode, VCF Operations registration does not require manual transfer of files, the registration is performed by using an activation code. After the registration, usage reporting is automated, and the update of licenses is simplified.

CAUTION:

To use VCF Operations for licensing, verify that the Virtual Management Infrastructure (VIM) adapter is running.

By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.

If you do not want to include the FQDN in the registration details, you can use the following API:

```
PUT {VCF_Ops_URI}/suite-api/api/deployment/config/globalsettings/GENERATE_DEFAULT_LICENSE_MANAGEMENT_ASSET_NAME/false
```

How to view the detailed contents of your registration file

If your VCF Operations instance is in connected mode, the registration details are sent automatically by using a combination of query string parameters and API parameters. If your VCF Operations instance is in disconnected mode, you must manually download the registration information as a JSON Web Signed (JWS) file. The information provided to Broadcom for registration is equivalent between connected and disconnected mode, but sent in different formats.

Note:

You can view the registration file contents only in disconnected mode.

- From the VCF Operations instance, navigate to **License Management > Registration**, and click **Download Registration File**.
- Use a text editor to open the file, and copy the content.
- Paste the content of the registration file into a JWT decoding tool.

After the content is decoded, you can view the collected registration information.

Table 820: Registration File Content

Property	Description
model_version	File format version: 1.0.0.
asset_name	The FQDN of the VCF Operations instance. It is pre-populated as display name for the instance in the VCF Business Services console. This is only saved by Broadcom if you choose to keep the display name the same as the FQDN.
created_on	Date and time when the registration file was created.
asset_type	AC
xr2	A unique opaque fingerprint of the VCF Operations instance. It is used in conjunction with the asset_id to link subsequent usage data to the registered VCF Operations instance. No identifiable details about your environment can be derived from the fingerprint.
asset_id	The unique ID of the VCF Operations instance you register with this file.
request_id	Unique ID of the registration file.

Register VCF Operations in Connected Mode

You must register the VCF Operations instance that you want to use for license management with the VCF Business Services console. If your environment is connected to the Internet, you can register your VCF Operations instance in connected mode. Connected mode provides a faster and simplified registration process, and makes updating your licenses easier. Connected mode is recommended for anyone who has an Internet connection.

- Verify that you have a valid subscription for VCF or vSphere Foundation.
- Verify that your VCF Operations instance is connected to the Internet. If you need to configure an Internet proxy, see [Configuring Cloud Proxies in VCF Operations](#).
- Verify that you have either a user administrator or a product administrator role in your support site to access the VCF Business Services console. Alternatively, a user administrator must give you tenant access.

- Log in to VCF Operations, and select **License Management > Registration**.

- In the connected mode pane, click **Start Registration**.

The VCF Business Services console opens in a new tab.

- Log in to the VCF Business Services console by using your Broadcom Support Portal credentials.

The registration wizard appears.

- Select the Site ID to which you want to register this VCF Operations instance and click **Next**.

If you have only one site, it is selected by default.

- Optional: Enter a unique display name for your VCF Operations instance, and click **Save and Next**.

By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.

If you do not want to include the FQDN in the registration details, you can use the following API:

```
PUT {VCF_Ops_URI}/suite-api/api/deployment/config/globalsettings/GENERATE_DEFAULT_LICENSE_MANAGEMENT_ASSET_NAME/false
```


- Optional: Select licenses to add to your VCF Operations instance, and click **Save and Next**.

Note:

To license your environment, you must add a primary license to your VCF Operations instance. You can add licenses during the registration process, or at any time after that.

- View the summary of the registration, and click **Generate Activation Code**.

To complete the registration, you must copy the activation code and enter it in VCF Operations .

- Click **Copy**, and click **Finish**.

- In the VCF Operations instance, navigate to **License Management > Registration**.

- In the connected mode pane, click **Enter Activation Code**.

- Paste the activation code, and click **Activate**.

- If you failed to copy and paste the activation code, you must generate a new, valid activation code, and paste it in VCF Operations:

- In the VCF Operations instance, navigate to **License Management > Registration**.
- In the connected mode pane, click **Start Registration**.

The VCF Business Services console opens in a new tab.

- Log in to the VCF Business Services console by using your Broadcom Support Portal credentials.
- Select the Site ID to which you want to register this VCF Operations instance and click **Next**.

A new activation code is presented.

- Copy the activation code and click **Finish**.
- To paste the code in the VCF Operations instance, follow steps 8-10.

- Add licenses to your VCF Operations instance. For more information, see [Add a License to VCF Operations](#).
- Update the licenses at least once every 6 months. For more information about updating licenses, see [Update Licenses](#).

Register VCF Operations in Disconnected Mode

You must register the VCF Operations instance that you want to use for license management with the VCF Business Services console. If your environment is disconnected from the Internet, you can register your VCF Operations instance in disconnected mode.

- Verify that you have a VCF Operations instance version 9.0.
- Verify that you have a valid subscription for a VMware product.

After you register your VCF Operations instance in disconnected mode, you must manually submit usage reports in the VCF Business Services console. Then you must update your licenses by importing a license file in the VCF Operations instance.

- Log in to VCF Operations, and select **License Management > Registration**.
- In the disconnected mode pane, click **Download Registration File**, and save the registration file.
- Use a computer with Internet connection and log in to the VCF Business Services console (vcf.broadcom.com) with your Broadcom Support Portal credentials.

- Select the Site ID to which you want to register this VCF Operations instance.

If you have only one site, it is selected by default.

- Navigate to **License Management > VCF Operations Registrations**, and click **New Registration**.

The New Registration wizard appears.

- Click **Browse**, and upload the registration file that you downloaded from VCF Operations.

- Click **Save and Next**.

Note:

If you upload a registration file that is already used to register another VCF Operations instance, the operation fails with an error.

- Enter a unique display name for your VCF Operations instance, and click **Save and Next**.

By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.

If you do not want to include the FQDN in the registration details, you can use the following API:

```
PUT {VCF_Ops_URI}/suite-api/api/deployment/config/globalsettings/GENERATE_DEFAULT_LICENSE_MANAGEMENT_ASSET_NAME/false
```

- Select licenses to add to your VCF Operations instance, and click **Save and Next**.

- View the summary of the registration, and click **Save and Next**.

To complete the registration, you download a license file and upload it in VCF Operations.

- Download the license file, and click **Finish**.

- In the VCF Operations instance, navigate to **License Management > Registration**.

- In the disconnected mode pane, click **Import License File**.

- In the **Import License File** dialog box, click **Browse**, and select and import the license file.

- If you failed to download and import the license file, you can generate a new, valid license file, and import it in VCF Operations without registering again:

- In the VCF Business Services console console, navigate to **License Management > VCF Operations Registrations**.

- Next to the VCF Operations instance you want to register, click on the vertical ellipsis button.

- From the actions menu, select **Generate License File**.

A dialog box appears.

- When the license file is generated, download the file, and click **Finish**.
- To import the file in the VCF Operations instance, follow steps 12-14.

- Click **Complete**.

- Add licenses to your VCF Operations instance. For more information, see [Add a License to VCF Operations](#).
- Update the licenses at least once every 6 months. For more information about updating licenses, see [Update Licenses](#).

Update a VCF Operations Registration

If a change is made to your environment and it affects your ability to update your licenses, you must update the registration of the VCF Operations instance with the VCF Business Services console.

When this change occurs, a banner appears on top of the screen and requires you to update the VCF Operations registration. To be able to continue updating or importing your licenses in this VCF Operations instance, you must first update the registration.

1. In the banner, click Generate Updated Registration File.
2. Download the registration file.
3. Log in to the VCF Business Services console.
4. Navigate to **License Management > VCF Operations Registrations**.
5. Click the vertical ellipsis next to the VCF Operations instance that you must update.
6. Click **Upload Updated Registration File**.
A dialog box appears.
7. In the **Upload Updated Registration File** dialog box, click **Browse** and upload the file that you downloaded from your VCF Operations instance.
8. Select the attestation check box, and click **Import**.
9. Click **Close**.

The registration of the VCF Operations instance is updated. You can manage the licenses added to the instance.

Switch from Connected to Disconnected Mode

You can switch from connected to disconnected, and from disconnected to connected mode at any time. In connected mode, usage reporting is automated, and the update of licenses is simplified.

1. Log in to VCF Operations, and select **License Management > Registration**.
2. Click **Switch to Disconnected**.
3. In the **Switch Reporting Mode** dialog box, click **Switch to Disconnected**.

Your VCF Operations instance operates in disconnected mode. You must manually report license usage, and then upload license files in VCF Operations. For more information about reporting license usage and updating your licenses, see [Report License Usage and Update Licenses in Disconnected Mode](#).

Switch from Disconnected to Connected Mode

You can switch from connected to disconnected, and from disconnected to connected mode at any time.

To switch from disconnected to connected mode, verify that your environment is connected to the Internet.

If your VCF Operations instance operates in disconnected mode for more than 180 days, switching from disconnected to connected connection might fail, and you must generate a new activation code. For information on how to generate a new activation code, see [Register VCF Operations in Connected Mode](#).

1. Log in to VCF Operations, and select **License Management > Registration**.
2. Click **Switch to Connected**.
3. In the **Switch Reporting Mode** dialog box, click **Switch to Connected**.

Your VCF Operations instance operates in connected mode. License usage is reported automatically. You must update your licenses on regular intervals. For more information about updating your licenses, see [Update Licenses in Connected Mode](#).

Managing Licenses in the VCF Business Services console

Starting with version 9.0, you have one default license per product. You can split the default license into as many licenses as you need within your total purchased capacity. When you delete a split license, or reduce its capacity, the capacity is added back to the capacity of the default license.

User Roles

- Administrator roles from the site ID.

These allow you to add new users, you will have access to all tenants, you can manage licenses, etc.

1) Product Administrator

2) User Administrator

- Administrator roles from the tenant.

These roles are scoped to a given tenant. To grant a user these roles, they must have Site Access. 1) License Manager - "License Managers have the authority to perform operations with licenses. These operations include viewing, managing, and sharing licenses." 2) Tenant Admin - "Tenant Administrators have the authority to perform all service-related operations within the tenant. They can add new tenant members and manage all the resources of the tenant."

Split Default Licenses

You can split default licenses into as many licenses as you need within the total available capacity of the default license. You can allocate a part of the default license capacity to the new licenses that you create as a result of the split.

The total combined capacity of the split licenses cannot be larger than the total capacity of the default license.

1. In the VCF Business Services console, navigate to **License Management > Licenses**.
2. Click the **Version 9+** tab.
3. Click **Manage Licenses (Split/Merge)**.
The **Manage Licenses** wizard appears.
4. From the drop-down menu, select a product for which you want to split the default license.
5. Click **Next**.
The default license for this product appears.
6. Click **Create**.
A new line for the split license appears in the table.
7. In the **Name** column, enter a name for the license.
8. In the **New Capacity** column, enter the amount of capacity you want to allocate to the new license.
You cannot allocate higher amount of capacity than what is available in the default license.
9. Click **Next**.
10. On the Summary page, review your changes, and click **Finish**.

Your newly created license appears in the list of licenses in the VCF Business Services console.

If your default license is added to a VCF Operations instance, you must update the license in that VCF Operations instance after the split.

Merging Licenses

When you delete a split license, or reduce its capacity, the capacity you remove from the split license is merged back into the default license.

Change License Capacity

You can change the license capacity of split licenses as many times as you need. You can allocate more capacity or remove some of the capacity of the split licenses at any time. Any change in the capacity allocated to split licenses, affects the default license.

The capacity that you add to split licenses is extracted from the capacity of the default license. The capacity you remove from the split licenses is added back to the default license. You cannot add more capacity to a split license than the capacity available in the default license. If you need more capacity than what is available in the default license, you must free up capacity from the other split licenses which belong to the same product type, or you must purchase more capacity.

You cannot reduce the capacity of a license below the quantity of the license that is currently used. If you try to reduce the capacity of an assigned license, the operation fails. For example, if you have a license with capacity of 250 cores, and it is assigned to a vCenter instance, which has 4 ESX hosts added to it, each using 50 cores license capacity, you cannot reduce the capacity of the license to less than 200 cores.

Note:

If at a certain point the used capacity is higher than the capacity allocated to the assigned licenses, you must allocate more capacity to the licenses. Before you add more capacity, you cannot assign the over used licenses to new vCenter instances. If a vCenter is licensed by a license that is over used, you can only add new ESX hosts that are in evaluation mode.

1. In the VCF Business Services console, navigate to **License Management > Licenses**.
2. Click the vertical ellipsis next to the split license, and select **Change Capacity**.
3. In the **New Capacity** column, enter the new amount of allocated capacity.

For example, if your license has 150 cores of allocated capacity, and you want to add 50 cores, you add 200 in the **New Capacity** column.

You cannot assign a higher amount of capacity than what is available in the default license.

4. Click **Save and Next**.
5. On the Summary page, review your changes, and click **Finish**.

The new capacity of the license appears in the list of licenses.

Delete a License

You can delete a split license from the VCF Business Services console if you no longer need it. You cannot delete a split license if some of its capacity is assigned to a vCenter instance.

- Verify that the license is not assigned to a vCenter instance.
- Verify that the license is not added to a VCF Operations instance.

1. In the VCF Business Services console, navigate to **License Management > Licenses**.
2. Next to the split license, click the vertical ellipsis, and click **Delete**.
3. In the dialog box which appears, click **Delete**.

The license is deleted and its capacity is merged with the default license capacity.

Add a License to VCF Operations

To license your environment, you must first add licenses to the VCF Operations instance that you registered with the VCF Business Services console.

- Verify that you have a valid subscription for VCF or vSphere Foundation.
- Verify that your VCF Operations instance is registered with the VCF Business Services console.

After you add licenses to your VCF Operations instance, you can assign those licenses to the vCenter instances.

You can add any capacity to your VCF Operations, up to what you have purchased. A license cannot be added to more than one VCF Operations instance at a time.

1. In the VCF Business Services console, navigate to **License Management > Licenses**.
2. Click **Add to VCF Operations**.

The **Add Licenses to a VCF Operations Instance** wizard appears.

3. Select the licenses that you want to add, and click **Next**.
4. Select the VCF Operations instance to which you want to add the licenses, and click **Next**.
5. View the summary, and click **Save and Next**.
6. If your registration is in disconnected mode, download the license file, and click **Finish**.
7. To complete the process and have your licenses ready to assign to vCenter instances, update your licenses in the VCF Operations instance.

Option	Description
Registration in connected mode	Navigate to License Management > Registration , and click Update Licenses .
Registration in disconnected mode	1. Navigate to License Management > Registration. 2. Click Import License File. 3. In the Import License File pop-up window, click Browse. 4. Upload the license file, and click Complete.

The licenses are added to the VCF Operations instance. To view the license list in VCF Operations, navigate to **License Management > Licenses**, and select the **Version 9+** tab.

Assign a Primary License to a vCenter Instance

You assign a primary license to a vCenter instance. Your other assets that are connected to that vCenter instance, including ESX hosts, are then licensed automatically.

- Verify that you have the Manage Licenses permission assigned.
- Verify that the VCF Operations instance is registered with the VCF Business Services console.
- Verify that licenses are added to this VCF Operations instance.
- Verify that the vCenter instance to which you want to assign a license is integrated with the VCF Operations instance registered for license management. If the vCenter instance is part of a VCF domain, verify that the VCF Instance is integrated with the VCF Operations instance. For more information about adding vCenter instances and VCF instances to VCF Operations, see [Configuring a vCenter Account in VCF Operations](#), or [Configuring a VCF Cloud Account in VCF Operations](#).

Note:

After you integrate a vCenter instance or a VCF Instance with VCF Operations for license management, it takes up to 4 collection cycles before the vCenter instance appears in the **Licenses** tab of VCF Operations.

The vCenter instance can be standalone or part of a VCF Instance. You cannot assign a license to a vCenter instance if it requires more license capacity than the available capacity of the license you want to assign. If you try to assign the license to a vCenter instance which requires more capacity than the capacity available in the license, the assignment fails. If you try to add a host to a licensed vCenter instance, and the cores of the host are more than the available capacity left in the license assigned to the vCenter instance, the host is added in evaluation mode. If the host has exceeded the 90 day evaluation period, you cannot add it to the vCenter instance unless you increase the available capacity of the license.

Note:

If the host is in evaluation mode, it is added to the vCenter instance even if the license capacity is not sufficient. During the evaluation period of the host, you must add more capacity to the license to license the host. If the host is not licensed when the evaluation period expires, it is disconnected from the vCenter instance. The virtual machines running on the disconnected host remain intact. When there is enough available capacity in the license assigned to the vCenter instance, all disconnected hosts reconnect to the vCenter system.

For example, you add a license with 1000 cores allocated capacity to the VCF Operations instance. You then add 4 vCenter instances to be managed by this VCF Operations instance. The aggregate license consumption of the 4 vCenter instances requires 1500 cores. This means you cannot assign the license to all of them, as the aggregate consumption across all of the vCenter instances cannot exceed 1000 cores.

For more information about calculating the license capacity you need for your environment, see Broadcom knowledge base article [95927](#).

When a license expires, your hosts are disconnected from the vCenter instance, which can be a standalone instance or part of a VCF Instance, and you cannot start any workload operations. After you assign a valid license to the vCenter instance, the hosts are automatically reconnected to the vCenter instance unless you added them to another vCenter instance in the meantime.

1. In the VCF Operations instance, navigate to **License Management > Licenses**.
2. Select the **Version 9+** tab.

The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added the VCF Operations instance for monitoring only, it does not appear in this table.
3. From the vCenter Systems table, select one or more vCenter instances to which you want to assign a license.
4. Click **Assign Primary License**.

A list with the licenses added to VCF Operations appears. The licenses which you cannot assign appear at the bottom of the list, and the column **Assignable** shows No.
5. Select a license to assign, and click **Assign**.

In the **Primary License Name** column of the vCenter Systems table, you can view the licenses that you assigned to the vCenter instance. If the assignment failed for a vCenter instance, and you want to view the details of the task, click **View Details** in the banner above the table and hover over the information icon in the Status column.

Assign an Add-on License to a vCenter Instance

Add-on licenses are available for VCF and vSphere Foundation products. You can assign add-on licenses only after you assign primary licenses to the vCenter instances.

- Verify that you have the **Manage Licenses** permission assigned.
- Verify that your VCF Operations instance is registered with the VCF Business Services console.
- Verify that add-on licenses are added to the VCF Operations instance.

- Verify that the vCenter instance to which you want to assign a license is integrated with the VCF Operations instance. If the vCenter instance is part of a VCF domain, verify that the VCF Instance is integrated with the VCF Operations instance. For more information about adding vCenter instances and VCF Instances to VCF Operations, see [Configuring a vCenter Account in VCF Operations](#), or [Configuring a VCF Cloud Account in VCF Operations](#).

You cannot assign a license to a vCenter instance if that instance requires more license capacity than the available license capacity of the license you want to assign. To be able to use an add-on license, you must first add a primary license to your VCF Operations instance, and assign it to the vCenter instances.

1. In the VCF Operations instance, navigate to **License Management > Licenses**.
2. Select the **Version 9+** tab.
3. From the vCenter Systems table, select one or more vCenter instances to which you want to assign a license.

The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added to the VCF Operations instance for monitoring only, it does not appear in this table.
4. Click **Assign Add-on License**.

A list with the licenses added to VCF Operations appears. You cannot assign licenses which do not have the required capacity to license the selected vCenter instances. These licenses appear at the bottom of the list, and the column **Assignable** shows No.
5. Select a license to assign, and click **Assign**.

In the Add-on License Name column of the vCenter Systems table, you can view the licenses that you assigned to the vCenter instance. If the assignment failed for a vCenter instance, and you want to view the details of the task, click **View Details** in the banner above the table and hover over the information icon in the Status column.

Updating Licenses and Viewing the License Usage File

You must update your licenses in VCF Operations to verify that a usage file has been submitted, or when you make a license-related change such as subscription capacity changes, renewals, or splitting licenses and changing their capacity. Unlike prior VCF versions, when you update licenses, you do not need to reassign licenses to the assets.

If you do not make any license-related changes in the VCF Business Services console, you must still update your licenses at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations.

How to view the detailed contents of your usage file

You can view your license usage in a ready-to-read usage file that is decompressed and decoded, or you can view the raw usage file that is created in your VCF Operations instance. The raw usage file is JSON Web Signed (JWS) file compressed with gZip.

If your VCF Operations instance is in connected mode, usage files are automatically generated and submitted daily. If your VCF Operations instance is in disconnected mode, you must manually download and submit usage files on regular intervals. The content of the usage files are identical in connected and disconnected mode.

Note:

If your VCF Operations is in connected mode, you can only view the ready-to-read usage file.

View a ready-to-read usage file	1. In the VCF Operations instance, navigate to License Management > Registration. 2. Under File History, download the usage file you want to view, and open it.
---------------------------------	--

View the raw usage file in a human-readable format	1. Download and install gZip decompression software. 2. In the VCF Operations instance, navigate to License Management > Registration, and click Generate Usage File. 3. Use the decompression software and decompress the content of the usage file. The result is a file with the same name but no file extension. 4. Use a text editor to open the decompressed file, and copy the content. 5. Paste the content of the usage file into a JWT Decoding tool. After the content is decoded, you can view the collected license usage information.
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License Usage File Content

Property	Description
usage_report_generated_time	The date and time when the usage report is generated.
usage_report_id	The unique ID of this usage report.
asset_id	The unique ID of the VCF Operations instance where data is collected.
asset_status	<i>Active.</i>
xr2	A unique opaque fingerprint of the VCF Operations instance where data is collected; used in conjunction with the asset_id to link the usage data to the registered VCF Operations instance. No identifiable environment details can be derived from the fingerprint.
product_version	9.0 when the license is version 9+. 1.0 when a pre-version 9 license key is used.
product_name	Product display name.
Allocation_id	License ID for the version 9+ license.
license_key_id	License key for a pre-version 9 license.
usage_start_time	Start time of the usage period.
usage_end_time	End time of the usage period.
granularity	Always hourly.
uom	TiB or core for version 9+ licenses.
quantity	License usage quantity for this usage period.
total_quantity	License capacity (included for pre-version 9 licenses only).
unassigned_licenses	List of version 9 licenses (allocation_id) that are added to VCF Operations and are not used.
license_usage_anomaly_details	Generally empty. Indicates cases where usage data was corrupted - possibly as a result of a corrupt database, corrupt software update, or related circumstances.

The license usage file exclusively gathers this specific information and, for clarity, does not collect personal data or customer data.

Update Licenses in Connected Mode

After any change to your licenses in the VCF Business Services console, to reflect the changes in your environment, update your licenses in the VCF Operations instance. If your VCF Operations registration is in connected mode, the usage

file is submitted automatically on a daily basis. You must still update your licenses in the VCF Operations instance at least once every 180 days to verify that the usage file has been submitted correctly and to keep your environment licensed.

- Verify that you added one or more licenses to the VCF Operations instance.
- Verify that a usage file is generated after the last license update.

You must update your licenses at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations. You can also update your licenses at any time to extend your usage reporting refresh period.

1. In the VCF Operations instance, navigate to **License Management > Registration**.
2. To verify that your license usage file was submitted within the last 24 hours, check the **Last Usage Report Generated** date.
3. Click **Update Licenses**.

Your VCF Operations instance is updated with the latest license information available from the VCF Business Services console.

Report License Usage and Update Licenses in Disconnected Mode

When a VCF Operations instance is in disconnected mode, you must manually exchange the usage file and the license file between the VCF Operations instance and the VCF Business Services console.

- Verify that your VCF Operations instance is registered in the VCF Business Services console.

You must update your licenses and submit a license usage file at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations.

You can generate a new license file without submitting a usage file as long as it is within your usage reporting due date. As a best practice, always submit a usage file before you download a new license file.

1. In the VCF Operations instance, navigate to **License Management > Registration**.
2. Click **Generate Usage File**, and save the usage file.
3. In the VCF Business Services console, navigate to **License Management > VCF Operations Registrations**.
4. Next to the VCF Operations instance for which you want to update licenses, click the vertical ellipsis, and click **Upload Usage File**.
5. In the **Upload Usage File** wizard, upload the usage file, and click **Save and Next**.
A license file is generated.
6. Click **Download** and save the license file.
7. Click **Finish**.
8. In the VCF Operations instance, navigate to **License Management > Registration**.
9. Click **Import License File**.
The **Import License File** dialog box appears.

10. Click **Browse**, upload the license file, and click **Import**.

11. When the license file is imported, click **Complete**.

Remove a License from a vCenter Instance

To remove a license from a vCenter instance, you must assign a new license to the instance.

For more information about assigning a license to a vCenter instance, see [Assign a Primary License to a vCenter Instance or a VCF Instance](#).

Remove a License from VCF Operations

To remove a license from VCF Operations, the license must not be assigned to any vCenter instance in your environment.

- Verify that there are no licensed vCenter or VCF Instances managed by this VCF Operations instance.
- Remove all licenses from the vCenter instances. For more information, see [Remove a License from a vCenter Instance](#).
- Report license usage for the VCF Operations instance. For more information, see [Report License Usage and Update Licenses in Disconnected Mode](#).

After you remove the license from VCF Operations, you can assign the license to another VCF Operations instance or you can delete it. To complete the license removal, you must update your licenses in VCF Operations.

1. In the VCF Business Services console, navigate to **License Management > Licenses**.
2. Select the **Version 9+** tab.
3. Next to the license you want to remove, click the vertical ellipsis, and click **Remove License from VCF Operations**.

A dialog box appears.

4. Click **Remove**.
5. If your VCF Operations registration is in connected mode, update your licenses in the VCF Operations instance:
 - a) In the dialog box, click **Finish**.
 - b) In the VCF Operations instance, navigate to **License Management > Registration**.
 - c) Click **Update Licenses**.
6. If your VCF Operations registration is in disconnected mode, manually upload the license file in VCF Operations:
 - a) In the Download the License File dialog box, click **Download**, and save the license file.
 - b) Click **Finish**.
 - c) In the VCF Operations instance, navigate to **License Management > Registration**.
 - d) Click **Import License File**.
 - e) In the **Import License File** dialog box, click **Browse**, select and import the license file.
 - f) Click **Complete**.

To see the change in the VCF Operations instance, update your licenses. For more information about updating licenses, see [Update Licenses](#).

Deactivate a VCF Operations Instance

When you deactivate a VCF Operations instance from the VCF Business Services console, you can no longer add any licenses to that instance, and you cannot perform any license management operations.

- Verify that there are no licensed vCenter or VCF Instances managed by this VCF Operations instance.

- Remove all licenses from the VCF Operations instance. For more information, see [Remove a License from a vCenter Instance](#).
- Report license usage for the VCF Operations instance. For more information, see [Report License Usage and Update Licenses in Disconnected Mode](#).

Once you deactivate a VCF Operations instance, if you want to reactivate it, the same licenses added to it prior the deactivation might not be available.

1. In the VCF Business Services console, navigate to **License Management > VCF Operations Registrations**.
2. Click the vertical ellipsis next to VCF Operations instance you want to deactivate.
3. Click **Deactivate**.

A dialog box appears.

4. In the **Deactivate a VCF Operations Instance** dialog box, click **Deactivate**.

The VCF Operations instance is deactivated. You can view it in the list of VCF Operations Registrations with a status **Deactivated**.

Reactivate a VCF Operations Instance

You can reactivate a deactivated VCF Operations instance.

If you want to restore the same license configuration that was in use before the VCF Operations instance deactivation, add the same licenses to the instance that were added before the deactivation. If those licenses are not available, add new licenses to the instance.

1. In the VCF Business Services console, navigate to **License Management > VCF Operations Registrations**.
2. Click the VCF Operations instance that you want to reactivate.
The details and licenses page for the instance appears.
3. Click **Reactivate**.
A dialog box appears.
4. In the **Reactivate VCF Operations** dialog box, click **Reactivate**.
5. Add licenses to the VCF Operations instance. For information on how to add licenses, see [Add a License to VCF Operations](#).
6. Update your registration with the VCF Business Services console.

Option	Description
Connected Registration	1. In the VCF Business Services console, click the vertical ellipsis next to the VCF Operations instance. 2. Click Generate Activation code. 3. In the Generate Activation Code dialog box, copy the activation code, and click Finish. 4. In the VCF Operations instance, navigate to License Management > Registration. 5. Click Actions > Enter Activation Code. 6. In the Enter Activation Code dialog box, paste the activation code, and click Activate.
Disconnected Registration	1. In the VCF Operations instance, navigate to License Management > Registration, and click Generate Usage File.

Option	Description
	- In the VCF Business Services console, navigate to License Management > VCF Operations Registration, and click Upload Usage File. - In the Upload Usage File wizard, click Browse and upload the usage file that you generated in the VCF Operations instance. - Click Save and Next. A license file is generated. - Download the license file, and click Finish. - In the VCF Operations instance, navigate to License Management > Registration, and click Import License File. - In the Import License File dialog box, click Browse and import the license file. - Click Complete.

Your VCF Operations instance is reactivated and you can use it for license management. If you used the licenses that were previously added, you do not need to assign the license to the already licensed vCenter instances managed by this VCF Operations instance prior to its deactivation.

License Usage Analytics

License usage analytics are available both in the VCF Operations instance that you use for license management, and in the VCF Business Services console. You can access your license usage data at any time to check the aggregate license usage per product and per license.

With your subscription for a product, you also receive license keys for version 8.x of the product. You can decide which version of the product to deploy and license, but your total license usage must not exceed your purchased subscription capacity. You can use the usage analytics data to monitor license usage and change license capacity.

Note:

If at a certain point the used capacity is higher than the capacity allocated to the assigned licenses, you must allocate more capacity to the licenses. Before you add more capacity, you cannot assign the over used licenses to new vCenter instances. To the vCenter instances which are licensed with the over used licenses, you can only add new ESX hosts that are in evaluation mode.

View Product License Usage in the VCF Business Services console

In the VCF Business Services console, you can view aggregated data for your complete license usage for all your products and licenses in the unit of measure for the specific license. You can use the analytics data to monitor license usage and plan your license management.

You can also export and view the usage analytics data for product aggregate usage, version 9+ license usage, and pre-version 9 license usage. The export file is available in CSV and JSON formats.

- Log in to the VCF Business Services console (vcf.broadcom.com).
- Navigate to **License Management > Usage Analytics**.
- In the **Usage by Product** pane, click the product for which you want to view detailed license usage. The **Usage Details** page for the selected product appears.
- If you want to view usage analytics for a specific license, use the **Select License** drop-down menu.
- In the **Usage History** pane, you can view maximum license usage for a specific aggregation period.

Use the filters to view maximum license usage in monthly or daily periods.

View License Usage Per Product in VCF Operations

In your VCF Operations instance, you can view aggregated data for your license usage for products and for specific licenses, based on the licenses added to this VCF Operations instance. You can use the analytics data to monitor license usage and plan your license management.

- In the VCF Operations instance, navigate to **License Management > Usage Analytics**.
- In the **Usage by Product** pane, click the product for which you want to view detailed license usage. The **Usage Details** page for the selected product appears.
- If you want to view usage analytics for a specific license, use the **Select License** drop-down menu.
- If you want to view usage analytics for a specific vCenter instance, use the **vCenter** drop-down menu.

The list of vCenter instances includes only the instances to which you have assigned licenses from this VCF Operations instance. If you select a specific license to view data for, the list of vCenter instances contains only the vCenter instances to which this license is assigned.

- In the **Usage History** pane, you can view maximum license usage for a specific aggregation period.

Use the filters to view maximum license usage in monthly or daily periods.

License Management for Cloud Services Providers and Hyperscalers

Starting with version 9.0 of VMware Cloud Foundation (VCF) and VMware vSphere Foundation (vSphere Foundation), you license your environment by using a VCF Operations instance and the VMware Cloud Foundation Business Services console (vcf.broadcom.com). License files replace the use of the 25-character license keys.

Cloud Services Providers and Hyperscalers Requirements

To manage licenses as a provider (either cloud services provider or hyperscaler), you must meet the following requirements.

- Verify that you have an account in Broadcom Support Portal.
- Verify that you have an active commit contract with Broadcom.
- Verify that you have installed VCF Operations 9.0 and vCenter 9.0. The vCenter should be associated with VCF Operations.

What Is New?

- To license your environment, you must have entitlements to the VCF products.
- Once a commit contract is signed and available in the Broadcom Support Portal, you must create licenses against the contract serial number and the product in the contract.
- Depending on the commit contract, you have a different entitlement for the maximum license capacity. If the total allocated capacity in the created licenses reaches the maximum capacity, you can no longer create new licenses.
- End customers can share licenses for portability with you if you both meet certain requirements. For more information, see [License Sharing](#).
- To license your environment, you must have an installed VCF Operations instance, register it in the VCF Business Services console, and add licenses with available capacity. You can register VCF Operations to the VCF Business Services console either in connected or in disconnected mode. Connected registration is recommended for cloud services providers and hyperscalers.
- You must update your licenses at least once every six months. For more information, see [Updating Licenses](#).

- You assign licenses only to the vCenter instances. The other product components added to the licensed vCenter instances are then automatically licensed. For more information about products and product components, see [Products and Product Components](#).
- To collect the relevant product usage, you must deploy a VCF usage meter instance. VCF usage meter collects the license configurations in the metered environments and communicates the data to the cloud. The data is converted to core usage per hour for each license.
- You must register the VCF usage meter instance in the VCF Business Services console. The product usage data is available in the VCF Business Services console for partners to download and review.

License Management Between Tenants and Sites

You can manage licenses in all the tenants of the sites you belong to. However, you cannot transfer licenses between tenants. You must register the VCF Operations instance in the same tenant as the licenses you want to add. If you register the VCF Operations in a wrong tenant, you can deactivate the VCF Operations instance and restart the registration in the correct tenant. If you have added licenses, the licenses must first be removed before deactivating the VCF Operations instance.

Licensing Attributes for Providers

On the **Licensing** tab of the VCF Business Services console, you can see the attributes of the licenses you manage.

Licensing Attributes

Component	Description
Commit Capacity	The committed capacity in the signed commit contract. Can be in number of cores or TiBs.
Total Licenses	Total number of the created 9.0 licenses.
Consumed Capacity	The consumed capacity contains data for both non-running and running workloads. If VCF usage meter manages the environment on a running workload and the consumed capacity exceeds the commit capacity, you might get billed for over use.

High-Level Provider Licensing Workflow

- Install a VCF Operations instance version 9.0, or upgrade your existing Aria Operations instance to VCF Operations 9.0.
- Register your VCF Operations instance with the VCF Business Services console by using one of the two modes:
 - Disconnected mode - no Internet connection is required. You must manually exchange registration and license file between the VCF Operations instance and the VCF Business Services console. The file exchange must occur for every license adjustment and every 6 months at a minimum or the environment becomes out of compliance.
 - Connected mode - Internet connection is required. You use an activation code from the VCF Business Services console to register your VCF Operations instance.
- Add licenses to the VCF Operations instance from the VCF Business Services console, and update the license list in VCF Operations.
- Assign licenses to vCenter instances from VCF Operations.
- Deploy or upgrade a VCF usage meter appliance to version 9.0.

Create Licenses as a Provider

After you purchase a contract serial number and a product, you must create licenses against a contract serial number combination.

Verify that you are in the correct tenant.

Depending on the type of your contract, you can exceed the commit capacity for the created licenses. The capacity of the licenses must be between 1 and 100 000 cores.

- Log in to the VCF Business Services console.
- Navigate to **License Management > Licenses**.
- Click **Manage Licenses**.
- From the **All Serial Numbers and Products**, select the serial number and the product you want to create the new license against.

You can see all of the licenses previously created for the selected serial number and product with high-level details for commit contract capacity, the maximum capacity, total number of licenses created, and the total allocated capacity. You can also see the expiration date and consumed capacity of each license individually.

- Click **Create**.
- Enter a name and capacity for the license and click **Next**.
- View the summary and click **Finish**.

Register VCF Operations as a Provider

If the VCF Operations instance is provider managed, you should register it in connected mode.

Verify that you register the VCF Operations in the correct site and tenant.

The registered VCF Operations instance can be

- fully provider managed
- both customer and provider managed (applicable to white label customers)

If the VCF Operations is fully provider managed, you register the VCF Operations instance in connected or disconnected mode and manage all the licenses for the end customers within the tenant.

If both a customer and provider manage the VCF Operations instance, you must start the registration in disconnected mode and then pass the activation code to the end customer to enter into the VCF Operations instance. For more information, see [Register VCF Operations in Connected Mode](#) and [Register VCF Operations in Disconnected Mode](#).

Note:

When you add licenses in the VCF Business Services console, you can only select from existing licenses. Add only licenses created in the same tenant as the one you register the VCF Operations instance in.

Restart VCF Operations Registration

If you register the VCF Operations in the wrong site or tenant, you can restart the VCF Operations registration.

Before you restart the VCF Operations registration, you must remove any licenses added to the VCF Operations instance, delete the VCF Operations instance, and then re-register in the correct tenant.

- In VCF Operations, navigate to **License Management > Registration**.
- In the upper right corner, click **Actions**.
- Click **Restart Registration**.

License Sharing

License sharing has two forms - license sharing for portability and license sharing between tenants.

License Sharing for Portability

If the end customer purchases the VMware Cloud Foundation software from Broadcom, but buys the hardware, service and lifecycle management from a provider, the provider then deploys and manages the VMware Cloud Foundation environment for the end customer. To let cloud services providers and hyperscalers manage end customer licenses, end customers activate license sharing for portability on the licenses. End customers can then provide the license ID and the Site ID to a cloud services provider or a hyperscaler who validates the licenses. For more information, see [License Sharing for Portability](#).

License Sharing with Another Tenant

License sharing enables you to make a license in one tenant available for use in another tenant. For example, a company that does centralized purchasing, but has separate IT staff and deployments for each division of the company, can have a separate tenant for each division. Licenses purchased for a division can be shared from the purchasing tenant to the tenant for that division. If you want to share licenses between tenants, you must have access to both tenants. You can share a license from a default tenant where you have the **Tenant Administrator** or the **License Manager** role to any tenant where you have the **Tenant Administrator** or the **License Manager** role. For more information, see [Share a License with Another Tenant](#).

Terms Related to License Sharing

The following table describes terms that are relevant for license sharing.

Term	Use in License Sharing
Default License	A default license cannot be shared for portability or shared with another tenant. To make a license eligible for sharing, split the default license. For more information, refer to Licensing Terminology .
Split License	A split license is eligible for sharing for portability or for sharing with another tenant. If you want to share the license, verify it is not added to a VCF Operations instance. For more information, refer to Licensing Terminology .
Shared License	A license that has been shared for portability or with another tenant. As a result, adding the license to a VCF Operations instance can be performed in the tenant you share the license with. In the tenant you share the license from, the license appears in the Shared with Others tab. The license capacity can be modified in the tenant the license is shared from.

License Sharing for Portability

If an end customer purchases the VMware Cloud Foundation software from Broadcom, but purchases hardware, service and lifecycle management from a provider, the provider then deploys and manages the VMware Cloud Foundation environment for the end customer, which includes the license management. With license sharing for portability, cloud services providers can use the customer shared licenses to entitle a deployment.

License Sharing for Portability Requirements

The cloud services providers and hyperscalers must meet the following requirements.

- License management requirements for cloud services providers and hyperscalers. For more information, see [License Management for Cloud Services Providers and Hyperscalers](#).
- Certification for portability with the Broadcom Advantage Partner Program.

Licenses Eligible for Sharing for Portability

The following licenses are eligible for sharing.

- Primary licenses.
- Add-on licenses. The end customer must have a valid shared primary license as well.

For more information, see [Licensing Terminology](#).

High-level License Sharing for Portability Workflow

- The end customer creates a license of the desired capacity by splitting a default license.
- The end customer activates license sharing for portability for the newly created license.
- The end customer manually shares the license ID and the Site ID with the provider.
- The provider validates the license ID and site ID.
- Depending on who will manage the VCF Operations instance, the provider or/and end customer must have an installed VCF Operations instance and register it in the VCF Business Services console in the tenant of the end customer.
- The VCF Operations instance can be provider managed or customer managed.
 - If a provider manages the VCF Operations instances, the provider downloads a registration file for each of the VCF Operations instances in disconnected mode and passes the files to the end customer. The end customer then registers the VCF Operations instances in the tenant that contains the licenses they want to use for portability. After successful registration of the VCF Operations instances, the end customer receives a license file for each instance. The end customer must download the files and pass them back to the provider. The provider must upload the license files in the VCF Operations instances in disconnected mode. The provider can then switch to connected mode. For more information, see [Register VCF Operations as a Provider](#).
 - If the end customer manages the VCF Operations instances, after the provider gives the customer access to VCF Operations, the VCF Operations registration and license management are performed by the end customer. For more information, see [Register VCF Operations in Connected Mode](#) and [Register VCF Operations in Disconnected Mode](#).
- The provider deploys a VCF usage meter appliance and registers it in the VCF Business Services console for license usage reporting.

Activate License Sharing for Portability with a Provider

End customers can activate the sharing of a license for portability with a provider. The provider can then validate the license.

- Verify that the license type is **Split**.
- Verify that the license is not added to a VCF Operations instance.

After license sharing for portability for a license has been activated, the customer shares the license ID and site ID manually with the provider. The provider then validates the shared licenses for legitimacy and ownership by the end customer using the Broadcom API.

1. Log in to the VCF Business Services console (vcf.broadcom.com).
2. Under **License Management**, navigate to the **Vesion 9+ > Licenses** tab.
3. Click the vertical ellipsis next to the license.
4. Select **Manage Sharing for Portability**.
5. Activate the **Activate Sharing for Portability** toggle button.
6. Copy and send the License ID and Site ID to a provider.

After the cloud services provider or a hyperscaler completes validation of the license, you can add the license to a VCF Operations instance deployed by the provider.

Deactivate License Sharing for Portability with a Provider

If an end customer wants to share a license with a different provider or keep the license for personal use, they can deactivate license shared for portability.

Verify that the license is not added to a VCF Operations instance.

1. Log in to the VCF Business Services console (vcf.broadcom.com).
2. Under **License Management**, navigate to the **Vesion 9+ > Licenses** tab.
3. Click the vertical ellipsis next to the license.
4. Select **Manage Sharing for Portability**.
5. Click the **Activate Sharing for Portability** toggle button.

You receive a confirmation message. The license is no longer shared for portability with the provider.

Share a License with Another Tenant

You can share licenses from your default tenants to any tenant that you have access to.

Verify that the license type is Split.

You can perform the following set of actions in the source tenant.

- Increase capacity
- Decrease capacity (cannot be above the consumed capacity)
- Revoke the license sharing

You can perform the following set of actions in the destination tenant.

- Add to a VCF Operations instance
- Remove from a VCF Operations instance

Note:

Once you share the license with another tenant, the license is not eligible for splitting or merging.

1. Log in to the VCF Business Services console (vcf.broadcom.com).
2. Under **License Management**, navigate to **Version 9+ > Licenses**.
3. Click the vertical ellipsis next to the license.
4. Select **Share with Another Tenant**.

You receive the following message.

License *license-name* has been successfully shared with *tenant-name*. You can view the shared licenses in the Shared with Others tab.

The license now appears in the **Shared with Others** tab. In the **License History** tab, cloud services providers and hyperscalers can see the actions performed with the license before and after sharing.

Revoke License Sharing with Another Tenant

To share the license again or can manage the license in the tenant it was originally part of, you can revoke the license shared with another tenant.

Verify that the license is not added to a VCF Operations instance.

Note:

After you revoke the license successfully, the type of the license changes from **Shared** to **Split**.

1. Log in to the VCF Business Services console (vcf.broadcom.com).
2. Under **License Management**, navigate to **Version 9+ > Licenses**.
3. Navigate to the **Shared with Others** tab.
4. Click **Revoke Shared License**.

You receive a confirmation message. The license is no longer shared with another tenant.

The license is now part of the **Version 9+** tab as a **Split** license.

Move a Contract or a Serial Number Between Site IDs

Registration of VCF Operation is linked to a VCF Business Services console tenant and a Broadcom SiteID. The license allocations are linked to their contracts via serial numbers. If a contract or serial number is moved from one Broadcom SiteID to another, this impacts existing VCF Operations that are linked to the first SiteID with licenses from that SiteID.

Verify that you have a product or user administrator privileges for the new SiteID.

You cannot move serial numbers, you must contact support and request the move. In some cases, when you expand or renew a contract, and if a new SiteID is added in the ordering process, serial numbers might be moved.

If you need to move your serial number from one SiteID to another, you must first submit a usage report from VCF Operations and download a new license file. You need to add licenses again in the new site. You must do this for all VCF Operations instances with a license from that serial number.

If one of your licenses appears as **Inactive** but your subscription is not expired, your serial number might have been moved.

When a serial number is moved, you might observe the following:

1. Existing licenses linked to that serial number become **Inactive**.
2. Existing assignments appear as over assigned.
3. License updates fail.

4. After 180 days, your license expires, your hosts are disconnected from vCenter, and you cannot start workload operations.

After the serial number is moved, you must register your VCF Operations instance to the new SiteID and re-license your environment. Failure to do so may result in service impact to your environment.

1. Find the licenses and the VCF Operations instances related to the serial number.

- a) Log in to the VCF Business Services console.
- b) Navigate to **License Management > Licenses**.
- c) Use the search box to search for the serial number.

All of the filtered licenses contain the searched serial number. The search results will have a column for **VCF Operations Name**. These are the instances that will need to be re-linked to the new siteID.

2. Generate a license usage file from each VCF Operations instance related to the serial number, and update its licenses.

- a) Log in to the VCF Operations instance.
- b) Navigate to **License Management > Registration**.
- a) Update licenses.
 - If the VCF Operations instance is in connected mode, verify that your license usage file was submitted within the last 24 hours by looking at the **Last Usage Report Generated** date, and click **Update Licenses**. For more information, see [Update Licenses in Connected Mode](#).
 - If the VCF Operations instance is in disconnected mode, generate a license usage file. For more information, see [Report License Usage and Update Licenses in Disconnected Mode](#).

3. After you move your serial number, for each VCF Operations instance, create your licenses again by splitting the default license. For more information, see [Managing Licenses in the VCF Business Services console](#).

As a best practice, allocate the same capacity to the licenses as the capacity from the prior SiteID. Consider also using a distinct license name from the prior site.

4. Deactivate the VCF Operations instance in VCF Business Services console. For more information, see [Deactivate a Instance](#).

You must select the check box **I no longer have access to the VCF Operations instance**.

5. Restart registration for each VCF Operations instance linked to the serial number.

- a) Log in to the VCF Operations instance.
- b) Navigate to **License Management > Registration**.
- c) Click **Actions > Restart Registration**.

Important:

This action does not impact the existing licenses in your environment.

6. Register each VCF Operations instance linked to the serial number with the new SiteID. For more information, see [Registering VCF Operations with the VCF Business Services console](#).

CAUTION:

During registration, select the new Site ID, and the license you created in step 3.

7. Relicense your environment by assigning the licenses to the vCenter instances. For more information, see [Assign a Primary License to a Instance](#).

Both your old and new licenses are displayed in VCF Operations. After your next license update, if the old licenses are no longer assigned to vCenter instances, they do not appear in VCF Operations.

Update your licenses. For more information, see [Updating Licenses](#).